

Ziaul Haq Doost

Civil and Environmental Engineering

Water Resources and Environment Management Researcher

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RESEARCH PROFILE

Civil and environmental engineer with peer-reviewed research spanning hydrological modeling, water-resources planning, groundwater change, flood-frequency analysis, climate-change projections, GIS and remote sensing, and machine learning. Author or co-author of nine peer-reviewed publications, including five first-author studies. Experienced in taking research from problem formulation and data analysis through manuscript preparation, peer review, and publication, with additional field leadership in irrigation, drinking-water, sanitation, and community infrastructure projects.

RESEARCH INTERESTS

Hydrological and environmental modeling • Water-resources systems and reservoir planning • Climate-change impacts and adaptation • Flood-risk analysis • Groundwater and land-use change • GIS and remote sensing • Machine learning and uncertainty analysis • Water quality and treatment

EDUCATION

Master of Science in Civil and Environmental Engineering 2024

Water Resources Management and Environment | King Fahd University of
Petroleum and Minerals, Saudi Arabia

- Thesis: Runoff Management for Sustainable Water Supply in Harirud River Basin, Herat, Afghanistan.
- Integrated GIS-based multicriteria analysis with hydrological, environmental, geological, and socioeconomic criteria for reservoir siting and runoff-management planning.

Bachelor of Science in Civil Engineering 2015

Herat University, Afghanistan

Associate Degree Grade 14 in Construction Engineering 2011

Herat Technical Institute, Afghanistan

PEER REVIEWED PUBLICATIONS

1. Doost, Z. H., Yaseen, Z. M. **Allocation of reservoirs sites for runoff management towards sustainable water resources: Case study of Harirud River Basin, Afghanistan.** *Journal of Hydrology*, 634, 131042, 2024. doi:10.1016/j.jhydrol.2024.131042
2. Doost, Z. H., Yaseen, Z. M. **The impact of land use and land cover on groundwater fluctuations using remote sensing and geographical information system: Representative case study in Afghanistan.** *Environment, Development and Sustainability*, 27(4), 9515-9538, 2025; online 2023. doi:10.1007/s10668-023-04253-2

3. Doost, Z. H., Chowdhury, S., Al-Areeq, A. M., Tabash, I., Hassan, G., Rahnaward, H., Qaderi, A. R. **Development of intensity-duration-frequency curves for Herat, Afghanistan: Enhancing flood risk management and implications for infrastructure and safety.** *Natural Hazards*, 120(14), 12933-12965, 2024. [doi:10.1007/s11069-024-06730-x](https://doi.org/10.1007/s11069-024-06730-x)
4. Doost, Z. H., Alsuwaiyan, A., Abdulraheem, A., Al-Areeq, N. M., Yaseen, Z. M. **Rainfall prediction using integrated machine learning models with K-means clustering: A representative case study of Harirud Murghab Basin-Afghanistan.** *IEEE Access*, 13, 111628-111646, 2025. [doi:10.1109/ACCESS.2025.3581921](https://doi.org/10.1109/ACCESS.2025.3581921)
5. Doost, Z. H., Al-Areeq, A., Rahman, M., Chowdhury, S. **Predicting groundwater electrical conductivity with ensemble learning and clustering optimization.** *Environment, Development and Sustainability*, 1-38, 2026. [doi:10.1007/s10668-026-07532-w](https://doi.org/10.1007/s10668-026-07532-w)
6. Doost, Z. H., Goliatt, L., Aldlemy, M. S., Ali, M., Macedo, B. S. **Enhancing predictive accuracy of compressive strength in recycled concrete using advanced machine learning techniques with K-means clustering.** *AUIQ Technical Engineering Science*, 1(1), Article 10, 2024. [doi:10.70645/3078-3437.1009](https://doi.org/10.70645/3078-3437.1009)
7. Doost, Z. H., Alsuwaiyan, M., Yaseen, Z. M. **Runoff management-based water harvesting for better water-resources sustainability: A comprehensive review.** *Knowledge-Based Engineering and Sciences*, 5(1), 1-45, 2024. [doi:10.51526/kbes.2024.5.1.1-45](https://doi.org/10.51526/kbes.2024.5.1.1-45)
8. Yaseen, Z. M., Doost, Z. H., Simunek, J., Wan Mohtar, W. H. M., Homod, R. Z., et al. **Artificial Intelligence's Unresolved Challenges in Soil Heavy Metals Modeling: Proposing a Next-Generation Research Era.** *Earth Systems and Environment*, 1-8, 2026. [doi:10.1007/s41748-026-01278-1](https://doi.org/10.1007/s41748-026-01278-1)
9. Yaseen, Z. M., Doost, Z. H., Khan, R., Abdulraheem, A., Abdulameer, S. F., Falah, M. W., Farooque, A. A. **Chitosan-based flocculant heavy metal removal prediction using machine learning models.** *ACS Omega*, 10(40), 46714-46734, 2025. [doi:10.1021/acsomega.5c04000](https://doi.org/10.1021/acsomega.5c04000)
10. Aman, H., Doost, Z. H., Hejran, A. W., Mehr, A. D., Szczepanek, R., Gilja, G. **Survey on the challenges for achieving SDG 6: Clean water and sanitation: A global insight.** *Knowledge-Based Engineering and Sciences*, 5(3), 21-42, 2024. [doi:10.51526/kbes.2024.5.3.21-42](https://doi.org/10.51526/kbes.2024.5.3.21-42)
11. Yaseen, Z. M., Hameed, M. M., Doost, Z. H. **Validation of machine learning models for heavy metals bioavailability prediction: A comparative study.** *Journal of Environmental Chemical Engineering*, 13(3), 116749, 2025. [doi:10.1016/j.jece.2025.116749](https://doi.org/10.1016/j.jece.2025.116749)
12. Khan, R., Ilyas, S., Doost, Z. H. **Comparative assessment of land surface temperature in urbanized areas of Saudi Arabia.** *AUIQ Technical Engineering Science*, 2(1), Article 8, 2025. [doi:10.70645/3078-3437.1027](https://doi.org/10.70645/3078-3437.1027)
13. Nafea, A. A., Alawi, O. A., Doost, Z. H., AlAni, M. M., Ahmadullah, A. B., Deo, R. C., Yaseen, Z. M. **Electricity load and price forecasting in Spain: A hybrid deep learning framework leveraging temporal and seasonal dynamics.** *Expert Systems with Applications*, 299, 130123, 2026. [doi:10.1016/j.eswa.2025.130123](https://doi.org/10.1016/j.eswa.2025.130123)
14. Ahmadullah, A. B., Bakhtyal, I. A., Irshad, A. S., Doost, Z. H., Wali, N., Karimi, B. A. **Comparative assessment of optimal cleaning schedules for various PV cleaning techniques.** *Renewable Energy*, 256, 124182, 2026. [doi:10.1016/j.renene.2025.124182](https://doi.org/10.1016/j.renene.2025.124182)

15. Ahmadullah, A. B., Irshad, A. S., Doost, Z. H., Rahel, H. A., Qudah, A., Muslih, N. M., et al. **Techno-economic optimization of a hybrid heliostat-based CSP-hydropower-biopower system under hydro-limited conditions using Harris Hawks optimization.** *Scientific Reports*, 2026. doi:10.1038/s41598-026-60104-w
16. Ahmadullah, A. B., Rahimi, M. A., Ulfat, D. S., Irshad, A. S., Doost, Z. H., Wali, N., et al. **Decarbonizing Afghanistan: The most cost-effective renewable energy system for hydrogen production.** *Energy*, 324, 136000, 2025. doi:10.1016/j.energy.2025.136000
17. Ahmadullah, A. B., Noori, A. G., Alshammari, N. F., Irshad, A. S., Doost, Z. H. **Fueling the future with biomass-based green hydrogen: A case study from Afghanistan.** *Energy*, 139490, 2025. doi:10.1016/j.energy.2025.139490
18. Wali, N., Zaray, A. H., Ahmadullah, A. B., Doost, Z. H. **The impact of land use and land cover on land surface temperature and urban heat islands using geographical information system and remote sensing: A representative case study.** *Knowledge-Based Engineering and Sciences*, 6(3), 65-84, 2025. doi:10.51526/kbes.2025.6.3.65-84
19. Rasheed, N. J., Al-Khafaji, M. S., Alwan, I. A., Al-Suwaiyan, M. S., Doost, Z. H., et al. **Survey on the resolution and accuracy of input data validity for SWAT-based hydrological models.** *Heliyon*, 10(19), e38348, 2024. doi:10.1016/j.heliyon.2024.e38348
20. Hashim, B. M., Alnaemi, A. N., Hussain, B. A., Abduljabbar, S., Doost, Z. H., et al. **Statistical downscaling of future temperature and precipitation projections in Iraq under climate change scenarios.** *Physics and Chemistry of the Earth, Parts A/B/C*, 135, 103647, 2024. doi:10.1016/j.pce.2024.103647

RESEARCH AND TEACHING EXPERIENCE

Graduate Researcher and Teaching Assistant

2022–2024

Department of Civil and Environmental Engineering | KFUPM, Saudi Arabia

- Conducted MSc research on sustainable runoff management and reservoir-site allocation in the Harirud River Basin using GIS, multicriteria decision analysis, hydrological information, and infrastructure planning.
- Developed and contributed to peer-reviewed studies on groundwater response to land-use change, IDF curves and flood risk, water harvesting, climate downscaling, and hydrological-model input data.
- Prepared research datasets, performed quantitative analyses, interpreted results, developed figures and maps, and drafted and revised journal manuscripts through peer review.
- Supported undergraduate teaching through laboratory instruction, tutoring, grading, examination proctoring, course records, and preparation of instructional materials.

PROFESSIONAL ENGINEERING EXPERIENCE

District Engineer | MRRD Citizen Charter MCCG Program

2021

Afghanistan

- Directed technical surveys, designs, cost estimates, bills of quantities, environmental screening, environmental and social management plans, and construction supervision for community infrastructure subprojects.
- Reviewed technical proposals, as-built drawings, final estimates, monitoring findings, and management-information-system records for compliance, quality, sustainability, and cost control.

- Coordinated community, regional, and headquarters engineering stakeholders and supported corrective actions arising from third-party monitoring.

Project Manager | Tawana Services Development and Cultural National Organization **2020–2021**

Food for Assets Project | Herat, Afghanistan

- Managed project staff, schedules, beneficiary selection, stakeholder coordination, quality assurance, risk reporting, and monthly donor reporting.

Civil Engineer | OXFAM Novib **2017–2020**

Citizen's Charter National Priority Program | Herat, Afghanistan

- Supported community drinking-water, sanitation, and civil-infrastructure projects through survey, design, costing, proposal development, implementation monitoring, and quality control.

Civil Engineer | Shahran Construction and Engineering Ltd **2016–2017**

Irrigation Projects | Afghanistan

- Performed topographic and implementation surveys, engineering estimates, feasibility and cost-benefit analyses, cost control, reporting, and site assessments for irrigation and civil works.

TECHNICAL COMPETENCIES

Research and data: Python, MATLAB, statistical analysis, data visualization, machine learning, literature synthesis, academic writing, peer-review revision

Hydrology and environment: HEC-HMS, HEC-RAS, EPANET, hydrological analysis, flood-frequency analysis, groundwater assessment, water-resources planning

Geospatial and engineering: GIS, remote sensing, AutoCAD, Civil 3D, 3ds Max, engineering survey, quantity estimation, environmental and social safeguards

Productivity: Microsoft Word, Excel, Access, PowerPoint, and Outlook

SELECTED TRAINING

- Spatial Data Science The New Frontier in Analytics | Esri | 2022
- Solid Waste Management | Afghanistan with OXFAM Novib | 2018
- Engineering Construction of Decentralized Wastewater Treatment Systems | BORDA | 2015
- Water Resources Management Project | Asian Development Bank | 2014
- Construction Quality Control and Safety | USAID AWDP AMRAN | 2013

LANGUAGES

Dari — Native **Pashto** — Proficient **English** — Excellent